Vigil MD: final_v1 Release and CI

November 12, 2026 · Hitika Dalwadi · branch `model-dev/training-diagnostics` (commits 6afe449, 631be6b, 9d88a11) · PR #42 into `model-dev-integration` · follows *Vigil MD: Label and Feature Experiments* (November 10)

Summary

The experiments report recommended label_set_v2 (E1 + E3), feature set fs2 (F1 + F2), the IE gate (E4) and `xgb_no_match_kind`. We trained that recipe again as experiment **final_v1**. We packaged it with its TF-IDF encoder, cutoff and gate as one bundle and published it to GCP Artifact Registry as **matching-same-device v1.0.0**. On the hand sheet it reaches **precision 0.959 (3 false positives), recall 1.000**; on the test split, precision 0.9925 and recall 0.9654. We also fixed the model-training CI: smoke-train had been failing since fs2 became the default, and the train job now trains the released recipe. PR #42 is retargeted to model-dev-integration and its checks pass.

> **Status:** the model is released for **team testing** (MLflow alias `@challenger`), not for use. The 0.959 is optimistic: a 20-trial run of the same recipe scored 0.899, and the 10-trial recipe was kept after seeing both. Two things come before `@champion`: a blind second hand batch, and the chatbot-side code that finds candidates and builds the match context.

1. What was done

Step	What	Outcome
Train	s06 on `label_set_v2_fs2_features.parquet` (561,507 train rows, 22 features), experiment `final_v1`. The existing feature table was reused, so no new BigQuery cost	20 trials: 0.899. 10 trials: reproduces E3 exactly (section 2)
Evaluate	`eval/s08_all.py --ie-gate` on the hand sheet	xgb_no_match_kind 0.959 / 3 FP / recall 1.000
Serve	New `vigil_model/predict.py`: loads a bundle, builds features with the bundle's encoder, applies the IE gate before the cutoff	Same decision rule in scoring as in evaluation
Package	New `scripts/s11_package_final.py`: bundle, MLflow version and alias, manifest with sha256 of each file. Before writing, it checks the bundle reproduces s06's hand scores	Max score difference to s06 < 1e-5
Publish	Artifact Registry generic repo `vigil-md-models`, team naming convention	v1.0.0, downloaded back and checked
CI	Fixed smoke-train (fs2 TF-IDF encoder, fs2-only synthetic table). Train job on v2 + fs2 + gate	Green on PR #42; 509 tests pass
PR	PR #42 retargeted from same-device-profiling to model-dev-integration	Mergeable, no conflicts

2. Results of the final model

Hand sheet `hand_label_sample_v1`: 154 decided pairs (71 MATCH, 83 NON_MATCH) and 48 INSUFFICIENT_EVIDENCE (IE) pairs, scored with the IE gate at each model's validation cutoff (99% precision).

Run (label_set_v2, fs2, gate)	Precision (95% LB)	FP	Recall	IE accepted	Cutoff
xgb_no_match_kind, 10 trials (released)	**0.959** (0.899)	**3**	1.000	14 / 48	0.6179
xgb_no_match_kind, 20 trials	0.899 (0.825)	8	1.000	14 / 48	0.6172
xgb_all, 10 trials	0.899	8	1.000	14 / 48	0.6447
xgb_no_kind_derived, 10 trials	0.947 (0.882)	4	1.000	14 / 48	0.6368
logreg_v1	1.000 (0.956)	0	0.944	14 / 48	0.7940
E3 run (report, for comparison)	0.959	3	1.000	14 / 48	0.6179

Green: the released model. Red: the same recipe with 20 search trials. Test split of the released model: precision 0.9925, recall 0.9654, average precision 0.9982 (132,866 rows).

Why 20 trials did worse. The extra trials picked another hyperparameter setting with the same validation average precision (0.9980 against 0.9981). On the hand sheet it accepted 5 more wrong pairs: 7 of its 8 false positives are `contains` pairs and one is a placeholder-model pair. Validation is near its ceiling and cannot see these errors. This is the report's limit "model selection still uses the label-set validation split", now measured. The 10-trial run uses the same seed as E3 and reproduces it exactly (same cutoff, identical hand scores). Choosing it after seeing both results makes 0.959 optimistic.

3. The release

Naming follows the team convention: an MLflow registered model with auto-incremented versions and aliases; Artifact Registry versions by semantic version, cross-referenced to MLflow and git.

Where	Identifier	Value
Artifact Registry	repository / package	`vigil-md-1792696816 / us-central1 / vigil-md-models / matching-same-device` (generic format)
	version	`v1.0.0`: MAJOR for an input/output schema change, MINOR for features or architecture, PATCH for retraining
	annotations	`mlflow=mlflow-v1`, `git=git-6afe449`. Generic repositories reject tags (HTTP 501), so annotations hold them
MLflow	registered model	`matching-same-device` version 1, alias `@challenger`, tags `semver=1.0.0`, `git_sha`
	training history	every s06 run stays registered under `vigil-md-linkage-xgb_no_match_kind`
Git	serving code	commit `6afe449` contains `predict.py`. s11 refuses to package while the serving code is uncommitted

Bundle file	Content
`model.json`	XGBoost model (xgb_no_match_kind, 21 features)
`tfidf.pkl`	F1 encoder fitted by s04 on the training GUDID texts; the model learned its cosines from this vocabulary, so it must travel with the model
`meta.json`	cutoff 0.61790, feature columns, gate e6_e3b, MLflow run id, git commits, data and encoder sha256, hand-sheet results
`manifest.json`	package, version, cross-references, sha256 of each file (same layout as chatbot-query-router)
`README.txt`	how to load, decisions, limits

4. How a pair is decided

`Bundle.decide()` gives each pair a score and a decision. The gate runs first and the cutoff only decides the pairs the gate lets through:

- **INSUFFICIENT_EVIDENCE** if MAUDE has no usable model and no usable catalog (E6), or the identifiers are only prefix-related (E3b), whatever the score
- **MATCH** if the score is at least the cutoff (0.61790)
- **NON_MATCH** otherwise

```
gcloud artifacts generic download --project vigil-md-1792696816 --location us-central1 \
  --repository vigil-md-models --package matching-same-device --version v1.0.0 --destination models/v1.0.0

from vigil_model.predict import load
bundle = load("models/v1.0.0")
feats = bundle.features(maude_side, gudid_side, ctx)   # DeviceSide, DeviceSide, MatchContext
bundle.decide(pd.DataFrame([feats]))                # score + decision
```

5. CI pipeline

Problem	Cause	Fix
smoke-train failed at s05: `label_set_v1_features.tfidf.pkl not found`	Commit 20a5dad made fs2 the default. fs2 needs the TF-IDF encoder, which only s04 builds, and s04 needs GCP	`ci/make_ci_feature_table.py --tfidf` fits a small encoder on the committed hand sheet's GUDID texts; s05 reads it
Next failure behind it	The synthetic table held fs3 columns (incl. feat_labeler_sim), s05 built fs2	`--feature-set` (default fs2): the table holds only that set
Train job trained the old recipe	Hard-wired to label_set_v1, s08 without the gate	s03 to s06 on label_set_v2 + fs2, s08 `--ie-gate`, cache keyed on the v2 manifest, TF-IDF pickle uploaded with the models

Tests: 509 pass (3 new for CI, 2 for predict). The smoke job was reproduced locally and is green on PR #42. Not run yet: the GCP train job on v2. It runs only on a push to main or same-device-profiling, or by hand, and needs `label_set_v2` in the CI's BigQuery project.

6. Checks done

Check	Result
10-trial run against E3	Same cutoff (0.617902), hand scores identical line for line
Bundle against s06 (inside s11)	Max score difference below 1e-5 on all 202 hand pairs
Download from Artifact Registry	5 files; every sha256 matches the manifest
Downloaded bundle on the hand sheet	0.9595 / 3 FP / recall 1.000 / IE 14 of 48: same as at packaging
Unit tests and CI	509 passed; model-training and tests workflows green on PR #42

7. Limits

- **0.959 is optimistic.** The trial count was chosen after seeing the hand sheet, and the same recipe with 20 trials scored 0.899. The hand labels follow the labeling guide's rule tree, and with 83 NON_MATCH pairs, 2-3 false positives are within noise.
- **Known wrong links remain:** `contains` pairs with short numeric keys inside long catalog codes of the same labeler, and 14 of 48 IE pairs with agreeing identifiers are still accepted.
- **The bundle scores a given pair.** It does not find candidates, and it needs the match context (alias table, GUDID DI count per model, sibling keys), which today comes from BigQuery in s01 and s04.
- **tfidf.pkl is a pickle** and runs code when loaded: load bundles only from the team registry.

8. Next steps

- **Team access:** grant `roles/artifactregistry.reader` on vigil-md-models to testers.
- **Chatbot integration:** decide where candidate retrieval and the MatchContext inputs live (matching or chatbot), and how the chatbot installs `vigil_model`. Until then, testing is limited to pairs that already have features.
- **Run the CI train job once** by hand (workflow_dispatch) to confirm v2 in the CI project and the s04 cost.
- **Blind hand batch 2**, stratified by manufacturer size, renamed labelers, missing identifiers and contains cases. Use it to choose the trial count and the cutoff, then move the alias to `@champion`.
- Then the report's model work: E2 (TF-IDF neighbour negatives) for the contains collisions, F4 narrative identifiers, F5.

9. How to reproduce

```
cd Model-Development/matching
export VIGIL_EXP=final_v1
python scripts/s06_train_models.py --data data/label_set_v2_fs2_features.parquet \
    --manifest labels/label_set_v2.manifest.json --hand data/hand_features_v2_fs2.csv --trials 10
python eval/s08_all.py --exp final_v1 --ie-gate --hand-features data/hand_features_v2_fs2.csv
python scripts/s11_package_final.py --exp final_v1 --semver 1.0.0     # serving code committed first
AR="--project vigil-md-1792696816 --location us-central1 --repository vigil-md-models --package matching-same-device"
gcloud artifacts generic upload $AR --version v1.0.0 --source-directory dist/matching-same-device-v1.0.0
gcloud artifacts versions update v1.0.0 $AR --annotations=mlflow=mlflow-v1,git=git-6afe449
```

Run registry: `docs/EXPERIMENTS.md` (row final_v1). Release usage: README, section "Using the released model". Training logs: `reports/final_v1_train*.log` (local).